

Network Intrusion Detection using Classical AI and Machine Learning

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Abstract

Network intrusion detection is important for identifying malicious activities in computer networks. In this project, a simplified network traffic dataset containing 6000 records was used to classify connections as normal or attack. Different AI and machine learning techniques were applied including a Simple Reflex Agent, K-Nearest Neighbors (KNN), Naïve Bayes, and Logistic Regression. The results showed that machine learning models performed better than the rule-based reflex agent, with KNN giving the highest accuracy.

Introduction

A Network Intrusion Detection System (NIDS) is a security system that monitors network traffic and tries to detect suspicious activities or attacks. Since modern networks generate a large amount of traffic, it is difficult for administrators to manually inspect every connection. Machine learning can help automate this process by learning patterns from network data.

The goal of this project is to classify network connections as either normal or attack. Different AI techniques were implemented and compared. These techniques include a rule-based reflex agent, supervised learning algorithms, clustering, and a simple genetic algorithm for feature selection.

Dataset

The dataset used in this project is **network_traffic.csv**. It contains network connection records collected from a simplified intrusion detection dataset.

Dataset information:

Total records: 6000

Normal connections: 3000

Attack connections: 3000

Number of features: 15

Target column: label

Label 0 = Normal

Label 1 = Attack

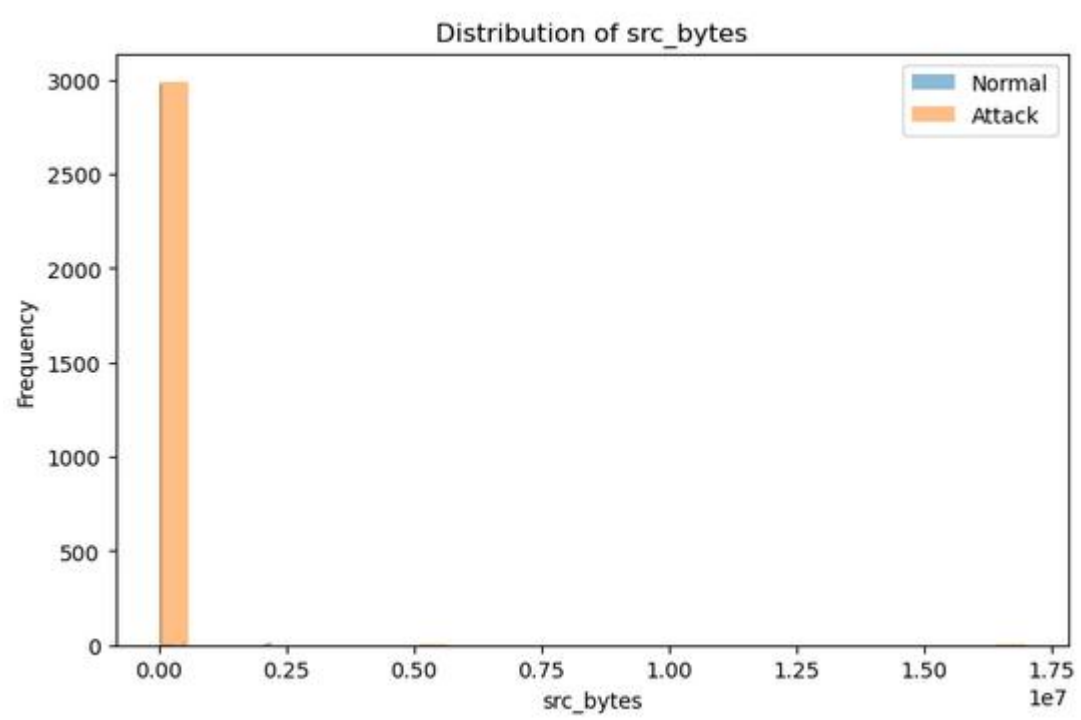
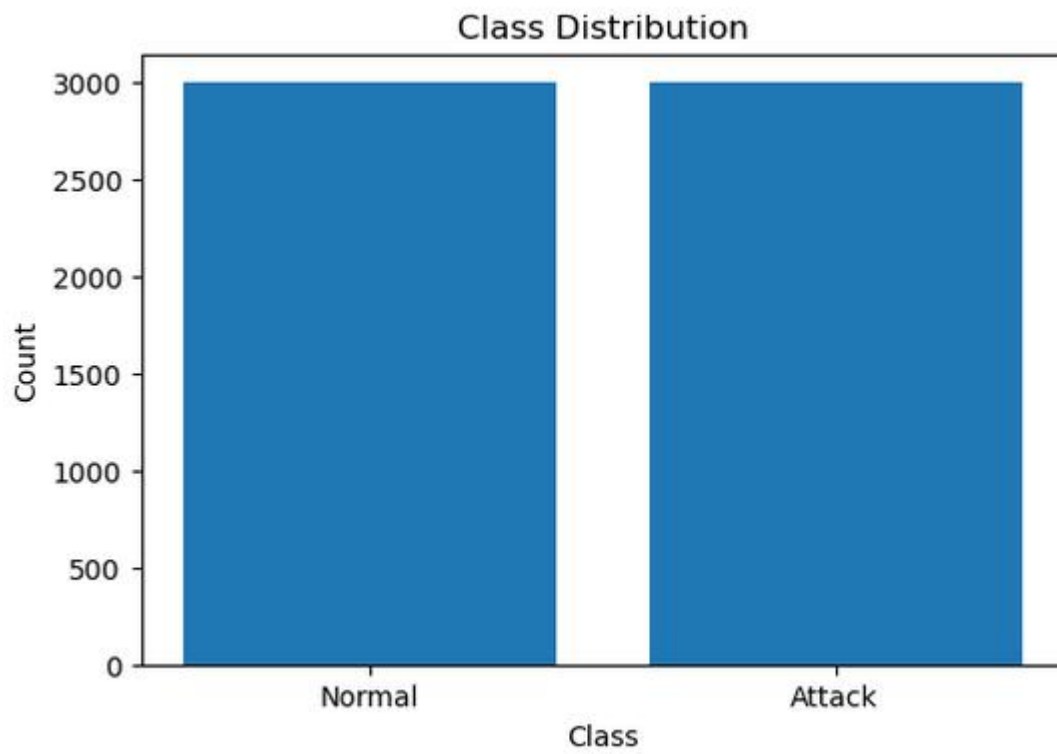
All features were already numeric, so no encoding was required. The dataset also had no missing values, which made preprocessing easier.

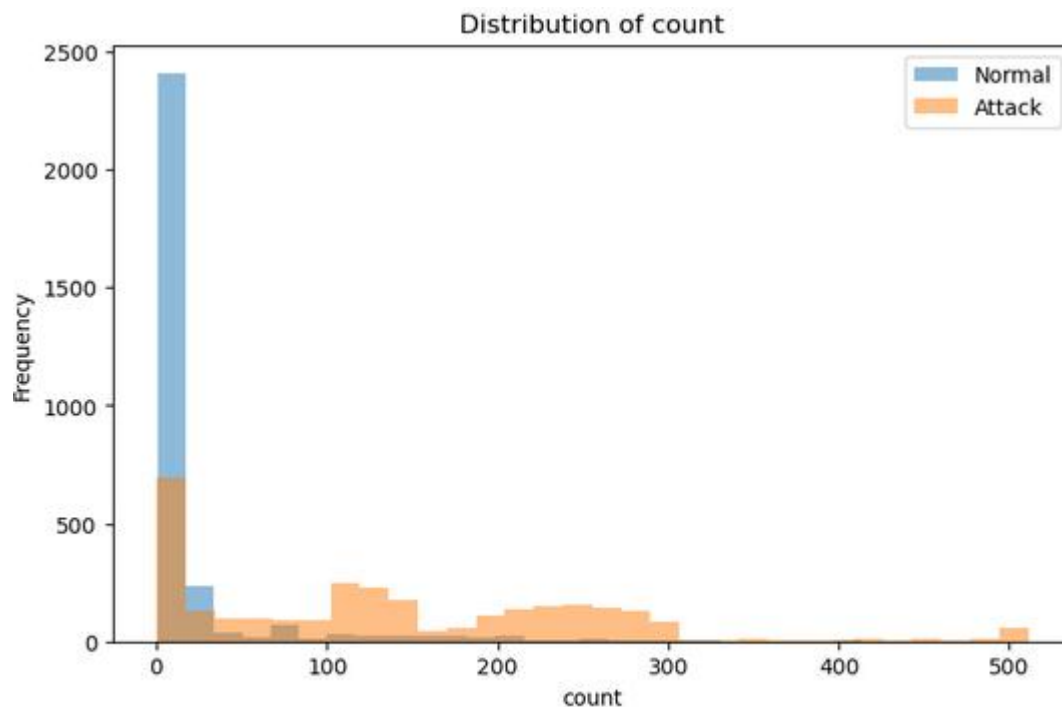
Task 1: Data Exploration

The dataset was first explored using pandas functions such as `head()`, `info()`, and `describe()`. These functions helped in understanding the structure of the data and feature values.

A bar chart was created to show the class distribution. The chart showed that the dataset is balanced because both classes contain 3000 samples.

Histograms were also plotted for features such as **src_bytes** and **count**. By comparing normal and attack traffic, some differences in feature distributions were observed. These observations helped in selecting rules for the reflex agent.





Task 2: Simple Reflex Agent

A simple reflex agent makes decisions using predefined if-then rules. It does not learn from data and does not store previous information. It only looks at the current input and takes an action.

The following rules were used:

If $\text{error_rate} > 0.5 \rightarrow \text{Attack}$

If $\text{count} > 100 \rightarrow \text{Attack}$

If $\text{same_srv_rate} < 0.4 \rightarrow \text{Attack}$

Otherwise $\rightarrow \text{Normal}$

Results:

Accuracy = 84.15%

Confusion Matrix:

```
[[2710 290]
 [661 2339]]
```

The reflex agent performed reasonably well, but it depends completely on manually selected rules. It may not work well if the traffic patterns change.

Task 3: Supervised Classifiers

The dataset was divided into training and testing sets using an 80:20 split. The same split was used for all classifiers to ensure a fair comparison.

Feature scaling was applied before using KNN and Logistic Regression because these algorithms are sensitive to differences in feature values.

Three classifiers were trained and tested:

Different values of K were tested: 1, 3, 5, 7, 9, and 11. The best result was obtained using the value that produced the highest test accuracy.

Gaussian Naïve Bayes was used because all features are numeric. This algorithm is simple and fast to train.

Logistic Regression was trained using scaled features and `max_iter = 1000`.

	Model	Accuracy	Precision	Recall	F1 Score
0	KNN	0.9575	0.949264	0.966667	0.957886
1	Naive Bayes	0.7875	0.936709	0.616667	0.743719
2	Logistic Regression	0.9150	0.924915	0.903333	0.913997

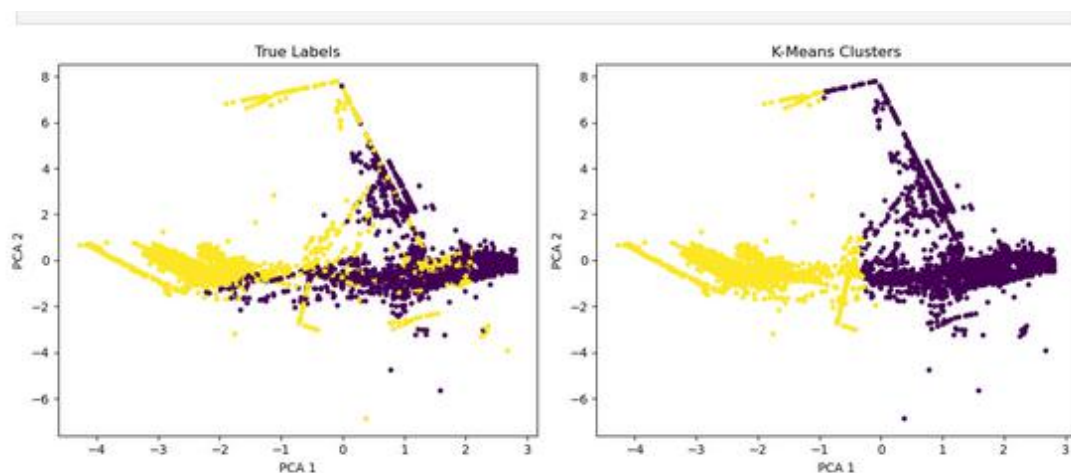
The results show that supervised learning models performed better than the rule-based approach. Logistic Regression achieved strong performance, while Naïve Bayes had lower accuracy compared to the other models.

Task 4: Clustering with K-Means

K-Means is an unsupervised learning algorithm that groups data into clusters based on similarity. In this task, the label column was removed and K-Means was used to create two clusters.

The generated clusters were compared with the actual labels using a confusion matrix. PCA was also used to reduce the data to two dimensions for visualization.

The scatter plots help show how the clusters relate to the real classes.



Task 5: Genetic Algorithm for Feature Selection

A simple Genetic Algorithm (GA) was implemented to find a good subset of features.

The chromosome was represented as a binary vector of length 15:

1 = feature selected

0 = feature not selected

The fitness value was the accuracy of Logistic Regression on the selected features.

The following GA operations were used:

Selection

One-point crossover

Bit-flip mutation

After several generations, the best feature subset was selected and compared with the model that used all features.

The reduced feature set required fewer inputs while maintaining similar performance.

Discussion

The reflex agent was able to classify many connections correctly, but its performance was limited because it relied on fixed rules. Machine learning models learned patterns directly from data and produced better results.

Among the supervised learning methods, KNN and Logistic Regression achieved strong performance. Naïve Bayes was faster but produced lower accuracy.

The genetic algorithm showed that a smaller feature subset can still produce good results. This can reduce model complexity and improve efficiency.

If more time were available, additional machine learning models and parameter tuning could be explored to further improve performance.

Conclusion

In this project, different AI techniques were applied to the network intrusion detection problem. Data exploration, rule-based classification, supervised learning, clustering, and feature selection were implemented successfully. The results showed that machine learning methods performed better than the simple reflex agent and can be effective for detecting network attacks.